Sahaj Bhadja

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Summary

Software engineer with 2+ years building autonomous-driving research pipelines, simulation systems, and reliable production services. Experienced in C++, Python, PyTorch, behavioral planning, trajectory processing, offline reinforcement learning, algorithms, Linux, testing, debugging, and performance analysis.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python APIs** and Azure SQL services for 500+ users, profiling indexed queries and backend execution paths to reduce production response latency from 1.2 seconds to 350 milliseconds.
- Implemented 5,000+ monthly workflows through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving observability, deterministic recovery, maintainability, operational resilience, and systematic production anomaly diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Flask REST services** handling 10,000+ daily requests, integrating secure document workflows, authentication, status tracking, and operational controls while systematically debugging production defects affecting 150+ users.
- Created **multithreaded processing** for 1,000+ files with validation, AWS RDS persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining reliable behavior under concurrent production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a GPT-based **Decision Transformer** for offline reinforcement learning, improving long-horizon autonomous-driving trajectory planning 18% in closed-loop simulation while developing end-to-end pipelines processing 100,000+ trajectory logs.
- Architected a spatial-temporal encoder integrating **HD-map geometry**, lane direction, ego state, and nearby-agent trajectories through multi-head attention, improving closed-loop planning stability 15% across diverse complex simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed scalable **Python services** and REST integrations using object-oriented patterns, AWS Glue, S3, SQL, filtering, pagination, and caching, improving production response times 30% and processing efficiency 68%.
- Maintained 20+ automated unit and integration suites with **Jenkins**, PyTest, and Mockito, accelerating defect investigation, release verification, and reliable deployments across Windows, Linux, and macOS production environments.

Pandit Deendayal Energy University **India**
Research Fellowship *January 2023 – May 2024*
- Designed a quantum-resistant blockchain architecture integrating five lattice-based cryptographic primitives, applying **distributed-systems and algorithmic reasoning** to strengthen protocol security against emerging computational threats within a research prototype.
- Analyzed consensus mechanisms and developed a novel algorithm reducing centralized-control and quantum-risk exposure 25% in benchmark simulations, contributing to a peer-reviewed technical paper presented at IISc Bengaluru.

Skills

Languages & Systems: C++, Python, Java, SQL, object-oriented programming, data structures, algorithms, operating systems, Linux, concurrency, multithreading, distributed systems
Planning & Machine Learning: behavioral planning, motion planning, trajectory processing, HD maps, geometry reasoning, agent interaction, simulation, offline reinforcement learning, Decision Transformers, PyTorch, multi-head attention
Data & Services: Flask, FastAPI, REST APIs, PostgreSQL, Azure SQL, AWS RDS, AWS Glue, S3, data pipelines, feature processing, cloud services, microservices
Testing & Engineering: Git, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, automated testing, debugging, profiling, performance analysis, root-cause investigation, validation, monitoring, documentation, Agile

Projects

CPU-Scheduling Simulator | Live | *C++, React, JavaScript, Algorithms, Operating Systems*
- Built a **C++ simulation system** implementing six scheduling algorithms with deterministic workloads, edge-case validation, Gantt visualization, and performance testing, reducing average turnaround time 25% across 1,000+ process configurations.

CaptionCraft | Live | *PyTorch, Transformers, Flask, Deep Learning, Streamlit, Gradio*
- Benchmarked **BLIP, GIT, and ViT-GPT-2** architectures using PyTorch on 10,000 images and 100,000 captions, improving prediction accuracy from 81% to 96% while building scalable real-time inference APIs.

Big-Brain | Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*
- Engineered a **data ingestion and vector-search pipeline** processing 50,000+ documents with incremental updates, metadata persistence, and performance tuning, reducing retrieval latency 40% while enabling contextual model inference.